

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of B. Pharm Examination

University Theory Examination April- 2016

MA131.1 Basic Mathematics

Date: 19/04/2016, Tuesday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION I is 30 minutes. Answers to SECTION I must be written during the first 30 minutes of the examination.
3. Answers to SECTION II and SECTION III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.
6. Use of scientific non-programable calculator is allowed.

SECTION I

Q.1. Darken the circle of the correct/most suitable choice. [20]

1. If $A = \{1, 3\}$ and $B = \{3, 4, 5\}$, $C = \{4, 5, 6, 7\}$ then $A \cap B =$

- (A) $\{1, 3\}$ (B) $\{3\}$ (C) $\{1, 4\}$ (D) $\{3, 4\}$

2. Suppose $n(U) = 70$, $n(A) = 30$, $n(B) = 45$ and $n(A \cap B) = 10$ then $n(A \cup B) =$

- (A) 5 (B) 25 (C) 65 (D) 20

3. The roster form of set $D = \{x : x \text{ is an integer, } -3 < x < 4\}$ is

- (A) $D = \{0, 1, 2, 3\}$ (C) $D = \{-3, -2, -1, 0, 1, 2, 3\}$
(B) $D = \{1, 2, 3\}$ (D) $D = \{-2, -1, 0, 1, 2, 3\}$

4. If u and v are differentiable function of x , a and b are constant,

then $\frac{d}{dx}(au + bv) =$

- (A) $a \frac{du}{dx} + c$ (C) $a \frac{d}{dx}(u) + b \frac{d}{dx}(v)$
(B) $a \frac{du}{dx} + \frac{dv}{dx}$ (D) None of these

5. Let $\int_0^3 f(x)dx = 5$ Then, $\int_3^0 f(x)dx =$

(A) 3

(B) 0

(C) -5

(D) 5

6. For the function $f(x) = x$, and for any value of x_0

(A) $\lim_{x \rightarrow x_0} f(x) = x_0$

(C) $\lim_{x \rightarrow x_0} f(x) = x$

(B) $\lim_{x \rightarrow x_0} f(x) = 1$

(D) $\lim_{x \rightarrow x_0} f(x) = 0$

7. If $f(x) = \cos x$ then $\frac{d}{dx}(\cos x) =$

(A) $\cos x$

(B) $-\cos x$

(C) $-\sin x$

(D) $\frac{1}{\sin x}$

8. Consider a universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and $A = \{1, 8, 9\}$,
 $B = \{1, 3, 4, 5, 6, 7\}$, $C = \{1\}$ then

(A) $C \subset B$

(B) $B \subset C$

(C) $A \subset C$

(D) $U \subset C$

9. If $f(x) = \sin x$, then $\int f(x) dx =$

(A) $\sin x$

(B) $\operatorname{cosec} x$

(C) $-\cos x$

(D) $\cos x$

10. for the function $f(x) = x^3$

(A) $\lim_{x \rightarrow 2} f(x) = 2$

(C) $\lim_{x \rightarrow 2} f(x) = 3$

(B) $\lim_{x \rightarrow 2} f(x) = 8$

(D) $\lim_{x \rightarrow 2} f(x) = 0$

11. The degree of polynomial $4x^3 - 6x + 2$ is

(A) 2

(B) 4

(C) 3

(D) 8

12. If $\begin{vmatrix} 4 & 1 \\ 1 & x \end{vmatrix} = 7$ then $x =$

(A) 2

(B) 3

(C) $-\frac{2}{3}$

(D) 7

13. For two matrices

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 3 \\ 4 & 2 \end{bmatrix}$$

then $2A + B =$

Ⓐ $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

Ⓒ $\begin{bmatrix} 3 & 5 \\ 7 & 6 \end{bmatrix}$

Ⓑ $\begin{bmatrix} 4 & 7 \\ 10 & 10 \end{bmatrix}$

Ⓓ $\begin{bmatrix} 2 & 3 \\ 4 & 2 \end{bmatrix}$

14. The matrix

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

is

Ⓐ diagonal

Ⓒ lower triangular

Ⓑ Identity

Ⓓ upper triangular

15. If

$$A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$$

then A'

Ⓐ $\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$

Ⓑ $\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$

Ⓒ $\begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}$

Ⓓ $\begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$

16. Suppose we have to solve system of two equations

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

Then x and y can be determined uniquely if

Ⓐ $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = 0$

Ⓒ $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = 1$

Ⓑ $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \neq 0$

Ⓓ $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = -1$

17. The following is a frequency distribution of blood pressure (mm/hg) of students.

Table 1: Class Limits

Lower Limit	60	63	66	69	72
Upper Limit	62	65	68	71	74
Frequency	5	18	42	27	8

The total number of students in table 1 are equal to

(A) 42

(B) 27

(C) 100

(D) 60

18. The median of 5 4 6 10 4 5 6 is

(A) 10

(B) 5

(C) 6

(D) 4

19. The harmonic mean of 2,4,and 8 is

(A) $\frac{7}{8}$

(B) $\frac{24}{7}$

(C) $\frac{8}{7}$

(D) $\frac{21}{8}$

20. The mode of numbers 45,36,28,42,45,40,44 is

(A) 42

(B) 45

(C) 44

(D) 40

SECTION II

Q.2. Attempt any **Four** of the following. [20]

A 1. Plot the function $f : \mathbb{N} \rightarrow \mathbb{R}$ define by $f(x) = x^2 - 1$. [02]

2. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $A = \{1, 2, 3, 4, 5\}$ and $B = \{4, 5, 6, 7\}$.

(a) Find $(A \cup B)$ (b) Show that $n(A \cup B)' = n(A' \cap B')$

where $n(A)$ = number element in set A, U is universal set [03]

B 1. Evaluate the limit $\lim_{x \rightarrow 5} 3x^4 + 3x^3 + 2x - 9$ [02]

2. Let $A = \{a, b, c, d, e, f, g\}$, $B = \{a, d, f, h, k\}$ and $C = \{d, e, h, k, l\}$.

Then show that $(A \cap B) \cap C = A \cap (B \cap C)$ [03]

C Evaluate the limit $\lim_{x \rightarrow 7} \frac{x^3 - 7^3}{x - 7}$ [05]

D Find first derivative of $f(x) = x^3 \sin x$ [05]

E Evaluate the limit $\lim_{x \rightarrow 0} (x^2 + 5x + \cos x - 2 \sin 3x + \sec x)$ [05]

F Examine the continuity of function $f(x) = e^x$ at $x = 0$. [05]

SECTION III

Q.3. Attempt any **Four** of the following. [20]

A 1. Define (i) Scalar Matrix (ii) Upper triangular matrix (iii) Symmetric Matrix [03]

2. Find $\int_1^3 (3x^2 + e^x) dx$ [02]

B Consider the matrices [05]

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 4 & 0 \\ 6 & 7 & 9 \end{bmatrix}$$

Find

(i) A' (ii) B' (iii) Show that $(A + B)' = A' + B'$

C 1. Evaluate $\int \frac{x}{5x^2 - 3} dx$ [02]

2. Find the determinant of a matrix D where [03]

$$D = \begin{bmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{bmatrix}$$

D 1. Evaluate $\int_0^5 (x^2 + x + 1) dx$ [03]

2. If

$$A = \begin{bmatrix} -1 & 1 \\ 3 & -1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$$

Find Matrix product AB

[02]

E Find the rank of matrix A where

[05]

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 5 & 4 & 9 \\ 6 & 2 & 0 \end{bmatrix}$$

F Evaluate $\int_0^1 \sqrt{2x+1} dx$

[05]

Q.4. Attempt any Four of the following.

[20]

A 1. Solve the quadratic equation $5x^2 - 2x - 4 = 0$

[03]

2. Define Median of variable

[02]

B Find the median of the following data:

[05]

Protein intake/day	15-25	25-35	35-45	45-55	55-65	65-75	75-85
Number of families	30	40	100	110	80	30	10

C The figures 14,13,11,10,7,9,8,12,10 show incubation periods of 9 polio cases.

Find Mean and Standard deviation of incubation period.

[05]

D Find Quartile deviation and coefficient of

Quartile deviation from the following data

[05]

Table 2: Marks of Student

Candidate No.	1	2	3	4	5	6	7
Marks	20	28	40	12	30	15	50

E Find the Arithmetic Mean of the following distribution:

[05]

Age	20-25	25-30	30-35	35-40	40-45	45-50
No. of Persons	170	110	80	45	40	35

F Solve using Cramer's rule $x + 2y = 9$, $2x - 3y = 4$

[05]